

CLINICAL INVESTIGATION

Targeted Tissue Perfusion Versus Macrocirculatory-Guided Standard Care in Patients With Septic Shock: A Randomized Clinical Trial—The TARTARE-2S Trial

OBJECTIVES: To determine whether treatment targeting improving tissue perfusion while allowing lower than recommended blood pressure (targeted tissue perfusion [TTP]) improves outcome compared with mean arterial pressure (MAP)-guided standard care (SC).

DESIGN: A randomized, parallel-group, open-label clinical trial with 30-day follow-up. Allocation was stratified according to trial site and presence of chronic arterial hypertension.

SETTING: Three European university hospital ICUs with enrollment from 2016 to 2022.

PATIENTS: Two hundred nineteen patients with septic shock and blood lactate greater than 3 mmol/L allocated to TTP ($n = 111$) vs. SC ($n = 108$).

INTERVENTIONS: The TTP protocol comprised capillary refill time, peripheral skin temperature, arterial lactate concentrations, and MAP 50–65 mm Hg. The SC protocol comprised the hemodynamic targets of 2012 Surviving Sepsis Campaign.

MEASUREMENTS AND MAIN RESULTS: Ninety-seven (87.4%) in TTP and 97 (89.8%) in SC group (total $n = 194$) were analyzed for the primary outcome. The median (interquartile range) of days alive in 30 days with normal lactate and without vasopressor/inotropic drugs (primary outcome) was 23 (10–27) in TTP group and 22 (1–27) in SC group (difference in medians, 0.59; 95% CI, –3 to 4). Secondary outcomes (single components of the primary outcome, days alive without organ support and mortality, all at 30 d) and serious adverse reactions were not significantly different between intervention groups. At day 30, 24 patients (24.7%) in TTP group vs. 27 patients (27.8%) in SC group had died. MAP levels were lower in the TTP group vs. the SC group.

CONCLUSIONS: In ICU patients with septic shock and lactate greater than 3 mmol/L, targeting tissue perfusion and allowing lower than recommended MAP did not increase the number of days alive with normal lactate and without vasopressor/inotropic drugs at 30 days. No additional safety concerns with the TTP strategy were detected compared with SC.

KEYWORDS: critical illness; lactate; mortality; septic shock; tissue perfusion; vasopressor

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DOI: 10.1097/CCM.0000000000006899

In septic shock, hemodynamic treatment guidelines on fluids and vasopressors have remained relatively unchanged (1, 2). Currently, mean arterial pressure (MAP) is the main target, accompanied by indicators of tissue perfusion, such as blood lactate (3, 4). These recommendations are largely



KEY POINTS

Question: Does targeting improving tissue perfusion while allowing lower than recommended blood pressure (targeted tissue perfusion [TTP]) hastens time to resolution of septic shock compared with blood pressure-guided standard care (SC).

Findings: This randomized clinical trial did not find a statistically significant difference in days alive in 30 days with normal lactate and without vasopressor/inotropic drugs: 23 days (10–27 d) in TTP group vs. 22 days (1–27 d) in SC group.

Meaning: The TTP strategy was feasible without additional safety concerns as compared with SC.

based on uncertain evidence, with the exception of MAP (moderate quality evidence).

While a recent randomized clinical trial (RCT) using tissue perfusion-oriented treatment strategies rather than MAP alone did not find differences in outcomes (5), a post hoc Bayesian analysis suggested high probability of benefit with a tissue perfusion-oriented treatment (6). Furthermore, lactate clearance as a surrogate of shock resolution has been associated with less days with organ failure and decreased mortality (7, 8). Surviving Sepsis guidelines recognize the lack of survival benefit from higher MAP targets (2–4). There are only three RCTs on effects of MAP targets on outcome in septic shock in ICU (9–11). Targeting MAP of 60–65 mm Hg instead of 75–80 mm Hg was associated with reduced mortality in a subgroup of patients older than 75 years in a pilot trial (10). In the subsequent large trial comparing MAP target of 60–65 mm Hg with usual care in vasodilatory hypotension, targeting 60–65 mm Hg did not result in reduction of mortality (11). Accordingly, the effect of MAP target on mortality remains uncertain. Vasoconstrictors for achieving MAP targets in septic shock may worsen tissue perfusion. Whether allowing lower MAP while targeting clinical signs of tissue perfusion may be beneficial is not known.

In this multicenter, phase II RCT in septic shock patients, we compared the effect of aiming at improving tissue perfusion while accepting lower MAP (targeted tissue perfusion [TTP]-guided treatment

strategy) with MAP-guided standard care (SC) on the number of days alive with normal lactate and without vasopressors or inotropes in 30 days.

METHODS

Trial Design

We performed a multicenter, investigator-initiated, randomized, open-label, parallel-group trial to compare the effect of TTP with SC in 219 septic shock patients. The trial protocol was published (12) and approved by the ethics committees at each trial site and the European Medicines Agency (Targeted Tissue Perfusion Versus Macrocirculatory-Guided Standard Care in Patients With Septic Shock; 10192015). The procedures were followed in accordance with the ethical standards of the responsible committees on human experimentation and with the Helsinki Declaration of 1975 (further details in **Electronic Supplementary Materials**, <https://links.lww.com/CCM/H815>).

We screened the patients for inclusion and exclusion criteria. Due to the urgency of septic shock, the ethics committees provided a possibility of a deferred consent policy, if the patient was unable to consent and the proxies were not available. In case the patient was unable to provide consent, we obtained a proxy consent from next of kin before randomization. In addition, in Switzerland and Denmark, before randomization an approval from an independent physician not involved with the care of the patient was mandatory. A deferred consent from the patient was obtained as soon as possible. If consent was withdrawn or withheld, we asked permission to use data for the primary outcome and/or data collected until consent refusal or withdrawal. An independent data and safety monitoring board reviewed the outcomes during the study.

Patient Population

We screened ICU patients with septic shock between June 9, 2016, and May 8, 2022, at three hospitals (one in Switzerland, one in Finland, and one in Denmark). Eligible patients were 18–80 years old, had septic shock defined as: 1) infection (suspected or documented); 2) need of any dose of vasopressors (norepinephrine, epinephrine, vasopressin) to achieve MAP greater than or equal to 65 mm Hg despite adequate fluid resuscitation (a minimum of 20 mL crystalloids/kg actual body

weight within 2 hr to facilitate early recruitment); and 3) an arterial lactate of greater than or equal to 3.0 mmol/L with presumed hypoperfusion, as judged clinically. We selected a lactate concentration threshold of greater than or equal to 3.0 mmol/L to exclude patients with transient, mild lactate elevations and to ensure the inclusion of septic shock patients with more severe illness. All inclusion criteria had to be fulfilled within 4 hours of ICU admission and the patient randomized within 4 hours of fulfilling the inclusion criteria. Patients were excluded, if any other condition than sepsis was expected to affect their CNS, including post cardiac arrest, if they had myocardial ischemia, acute pulmonary embolism, severe liver, and/or chronic kidney disease or a terminal illness without considering full intensive care, were on extracorporeal membrane oxygenation, or had another probable cause of hyperlactatemia (details in Electronic Supplementary Materials, <https://links.lww.com/CCM/H815>).

Randomization

We randomized patients within 4 hours fulfilling inclusion criteria. Using a computer-based algorithm created by an independent statistician to allow immediate allocation of patients in a 1:1 ratio to the intervention and control arms. Patients were stratified according to trial site and presence of chronic arterial hypertension, documented by specific medication in medical records and/or information from relatives. Varying block sizes, unknown to investigators, were used.

Intervention

The patients were treated using the targets of the allocated group. The treatment was continued until lactate was less than 2 mmol/L and no vasoactive drugs were used. Briefly, the TTP targets were: capillary refill time less than 3 seconds, absence of skin mottling, arterial lactate less than 2 mmol/L, warm peripheral skin temperature, urine output greater than or equal to 0.5 mL/kg/hr, and MAP 50–65 mm Hg (as a safety limit). The targets were chosen to reflect clinical resolution of shock. If patients had chronic hypertension, the MAP target was 65–70 mm Hg. In case of oliguria, we recommended a 2-hour trial of MAP of 75–80 mm Hg. If increase in diuresis followed, the higher MAP target continued and was reevaluated after each 2 hours. The hemodynamic problems (e.g., hypovolemia, impaired

contractility, etc.) were diagnosed by the treating clinicians—as interventions decided to correct them—similar to the previous RCTs on hemodynamic targets in septic shock (9–11). To assess hemodynamic problems, both echocardiography and pulmonary artery catheters were available and used at the discretion of the clinician in charge of the patient care.

Controls

The treatment using allocated targets was continued until lactate was less than 2 mmol/L and no vasoactive drugs were used. In SC, the primary targets were MAP of 65–75 mm Hg (if chronic hypertension 75–80 mm Hg), adequate fluid therapy to restore clinical hypovolemia, using central venous pressure (CVP) level of 8–12 mm Hg as safety limit. The urine output target was the same as in TTP. In case of oliguria, we recommended a 2-hour trial of MAP of 85–90 mm Hg. If diuresis increased, the higher MAP target was continued and reevaluated after each 2 hours. The hemodynamic problems occurring were diagnosed by the treating clinicians—as interventions decided to correct them.

There was no specific treatment protocol in either group, but only general treatment suggestions (Electronic Supplementary Materials, <https://links.lww.com/CCM/H815>). When the clinicians detected a hemodynamic problem, they decided about the given intervention/s. In both treatment arms, the respective treatment was supposed to lead to achievement of the unmet target/s. The list of possible hemodynamic problems and the list of available treatments were the same for both groups.

Data Collection and Monitoring

We collected data on baseline characteristics, physiologic variables, interventions, and days 30 and 90 outcome data from medical records including ICU patient data management systems, and from National Registers, using eCRFs. Lactate was measured in both groups every 2 hours until one confirmed value less than 2 mmol/L without vasopressors and inotropes or until 72 hours, whichever came first. In a subset of patients, the perfusion of kidneys, liver, and spleen was evaluated by ultrasonography (peak systolic velocity, end-diastolic velocity, resistive index, pulsatility index) at two time points, within 24 hours of ICU admission

and 24 hours after the first measurement, respectively. For logistic reasons, evaluation was only possible during weekdays. Ultrasonography was performed under the supervision of a certified radiologist following a standardized imaging protocol (13). Deaths up to day 90 and new acute kidney injury (AKI) were recorded (details in Electronic Supplementary Materials, <https://links.lww.com/CCM/H815>).

Outcomes

The primary outcome was the number of days alive with normal arterial blood lactate (first value < 2 mmol/L) and without inotropic agents or vasopressors at 30 days. No extra penalty was given for death. A day alive within 30 days and without vasoactive drugs and with normal lactate was counted as 1 day (24 hr from 00.00 to 24.00). Secondary outcomes were: time to normalization of lactate, number of days alive with normal lactate, days without inotropic or vasopressor agents, days without renal replacement therapy (RRT), days without mechanical ventilation, days without any organ support, all at 30 days, number of patients with new AKI (Kidney Disease: Improving Global Outcomes classification [14]), number of patients who received RRT, amount of norepinephrine given until day 5 (until 96 hr), and numbers of adverse and serious adverse reactions. All-cause mortality at day 90 was an exploratory outcome (details in Electronic Supplementary Materials, <https://links.lww.com/CCM/H815>).

Statistical Analysis

This was considered a phase II trial and, therefore, not powered for mortality or any of the secondary outcomes. Details about sample size calculation are presented in the Electronic Supplementary Materials (<https://links.lww.com/CCM/H815>). The study intervention and follow-up were stopped if the exclusion criteria were detected based on additional information after randomization. Thus, the analysis was based on a modified ITT principle.

We performed a Mann-Whitney *U* test for differences in continuous outcomes, and Fisher exact test for dichotomous variables as the primary analysis. We performed a subgroup analysis based on a past medical history of chronic arterial hypertension. A *p* value of less than 0.05 was considered statistically

significant. No imputations were performed due to low number of missing data (< 2% for all variables; details in Electronic Supplementary Materials, <https://links.lww.com/CCM/H815>). Treatment allocation was concealed from the trial statistician, the data and safety monitoring committee, and from the authors before analysis of the primary outcome.

RESULTS

Patients

During the patient recruitment from June 9, 2016, to May 8, 2022 (last follow-up, August 20, 2022, database closure, January 4, 2024),

Four thousand three hundred forty-three patients were assessed for septic shock, and 1298 had septic shock according to Sepsis-3 definition (3) (**Fig. 1**). We excluded 1079 of septic shock patients; the main exclusion criterion was lactate less than 3 mmol/L, and 219 (16.9% of 1298) septic shock patients were randomized. The recruitment was interrupted twice (2020, 2021) for a total of nine months due to COVID-19 pandemic. Nine patients were withdrawn due to delayed information on exclusion criteria and 16 patients excluded due to consent refusal or withdrawal, leaving 194 patients (97 in each group for the final analysis; **Fig. 1**). Baseline patient characteristics, including concomitant treatments in TTP and SC appeared similar (**Table 1**).

Trial Interventions

Treatment targets (**Tables S1 and S3**, <https://links.lww.com/CCM/H815>), achievement of the targets during the first 72 hours, until primary outcome, ICU discharge or death (**Tables S2 and S4 and Figs. S1–S9**, <https://links.lww.com/CCM/H815>), and selected interventions (**Table S5**, <https://links.lww.com/CCM/H815>) are presented in Electronic Supplementary Material (<https://links.lww.com/CCM/H815>). In 5181 records in TTP, the targets were achieved between 31% (lactate) and 72% (peripheral temperature). In 5222 records in SC, the targets were achieved between 43% (MAP) and 51% (CVP and urine output). Hemodynamic problems and respective treatments during the first 72 hours or until reaching the primary outcome are presented in **Tables S6 and S7** (<https://links.lww.com/CCM/H815>).

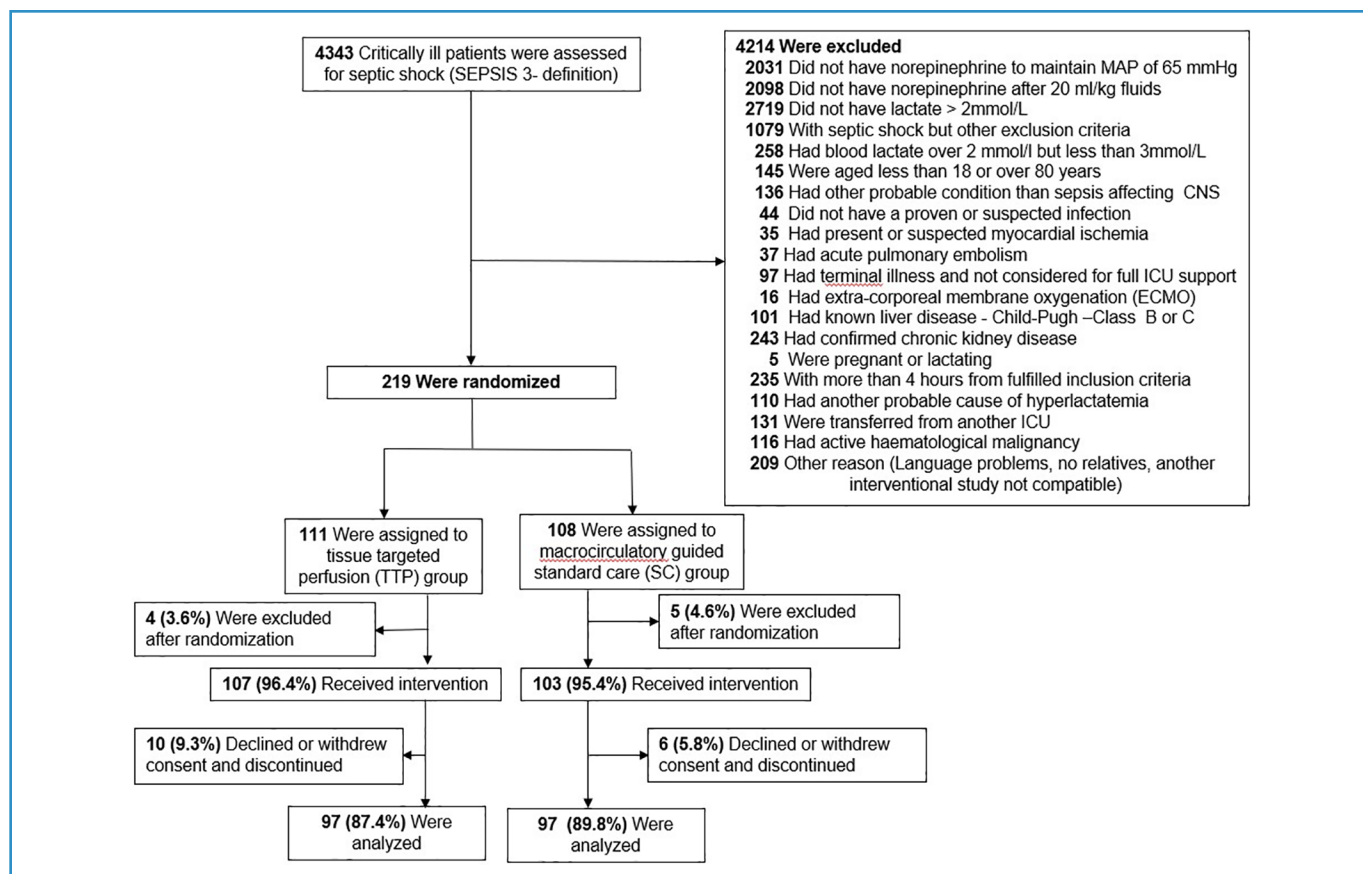


Figure 1. Trial flowchart. ECMO = extracorporeal membrane oxygenation, MAP = mean arterial pressure, SC = standard care, TTP = targeted tissue perfusion.

The MAP levels between 0 and 72 hours were lower in the TTP group vs. the SC group ($p < 0.001$; TTP: mean, 68.5 [SD, 11.4]; SC: mean, 74.8 [SD, 11.2]; mean difference [TTP vs. SC], -4.9 [95% CI, -7.0 to -2.9]; $p < 0.001$; **Fig. 2**), both among those with and those without previous hypertension (post hoc analysis; $p < 0.001$ for both comparisons; and **Figs. S10** and **S11**, <https://links.lww.com/CCM/H815>). Achievement of MAP targets between 0 and 72 hours, separately for patients with and without prior hypertension, is presented in **Table S8** (<https://links.lww.com/CCM/H815>). We performed MAP trials because of oliguria in 59 patients in TTP group vs. 57 patients in SC group. Post hoc analyses found a significant time-group interaction in norepinephrine infusion rates ($p < 0.001$; and **Fig. S12**, <https://links.lww.com/CCM/H815>).

We found no differences in cumulative fluid balance and fluid administration volumes between the groups (**Table S9**, <https://links.lww.com/CCM/H815>). The lactate levels ($p = 0.965$; **Fig. S13**, <https://links.lww.com/CCM/H815>) or numbers of patients with elevated

lactate while on vasopressors during the first 72 hours (**Fig. S14**, <https://links.lww.com/CCM/H815>) did not differ between groups. Kidney perfusion data among a subset of patients is presented in Additional Results (Electronic Supplementary Materials, <https://links.lww.com/CCM/H815>).

Primary Outcome

The median (interquartile range) number of days alive at 30 days with normal lactate and without vasopressor/inotropic drugs did not differ: 23 (10–27) in TTP group vs. 22 (1–27) in SC (difference in medians, 1; 95% CI, -3 to 4 d; **Table 2**). The distribution of number of days at 30 days with normal lactate and no vasopressors is shown (**Fig. 3**; and **Fig. S15**, <https://links.lww.com/CCM/H815>).

Secondary Outcomes

No statistically significant differences were found between groups in any component of primary

TABLE 1.
Patient and Treatment Characteristics at Baseline

Demographics, Diagnoses, Laboratory Values, and Baseline Treatment	Targeted Tissue Perfusion Group (n = 97)	Standard Care Group (n = 97)
Age, mean (sd), yr	62 (14)	62 (12)
Number of patients, n (%)		
Men	45 (46)	36 (38)
Women	52 (54)	60 (62)
Body mass index, mean (sd), kg/m ²	27 (7.8)	28 (6.2)
Concomitant disease, n (%)		
Hypertension	37 (38)	41 (42)
Chronic heart failure	7 (7)	10 (10)
Previous myocardial infarction	4 (4)	8 (8)
Previous stroke	5 (5)	6 (6)
Chronic obstructive pulmonary disease	8 (8)	10 (10)
Diabetes	21 (22)	18 (19)
Hematological malignancy	3 (3)	2 (2)
Metastatic cancer	6 (6)	7 (7)
Suspected or proven site of infection, n (%)		
Pulmonary	18 (19)	23 (24)
Abdominal	30 (31)	29 (30)
Urinary	1 (1)	2 (2)
Soft tissue	13 (13)	11 (11)
Other, including unknown	34 (35)	31 (32)
Baseline values, mean (sd)		
Creatinine, $\mu\text{mol/L}$, n = 105 ^a	81 (21)	81 (26)
Lactate, mmol/L	6.2 (3.2)	5.6 (2.8)
Hemoglobin, g/L	116 (20)	111 (26)
Mean arterial pressure	65 (9)	66 (10)
Treatment at randomization, n (%)		
Mechanical ventilation ^a	59 (61)	71 (73)
Renal replacement therapy ^a	21 (21.6)	23 (23.7)
Norepinephrine dose, $\mu\text{g/kg/min}$	0.2 (0.2)	0.2 (0.2)
Use of inotropic agents ^a	45 (46.4)	42 (43.3)
Fluid volume given—the last 2 hr, mL, mean (sd)	1100 (1033)	1129 (1061)
Simplified Acute Physiology Score II score, mean (sd)	57 (17)	55 (17)
Sequential Organ Failure Assessment score, mean (sd)	8.9 (2.5)	8.6 (2.4)

^aDuring the preceding 24 hr.

outcome. At day 30, 24 of 97 (24.7%) in TTP group vs. 27 of 97 patients (27.8%) in SC group had died (**Fig. S16**, <https://links.lww.com/CCM/H815>). Other secondary outcomes did not differ between groups (Table 2).

Exploratory Outcomes

No statistically significant differences between groups were found in exploratory outcomes or adverse reactions (**Table S10**, <https://links.lww.com/CCM/H815>).

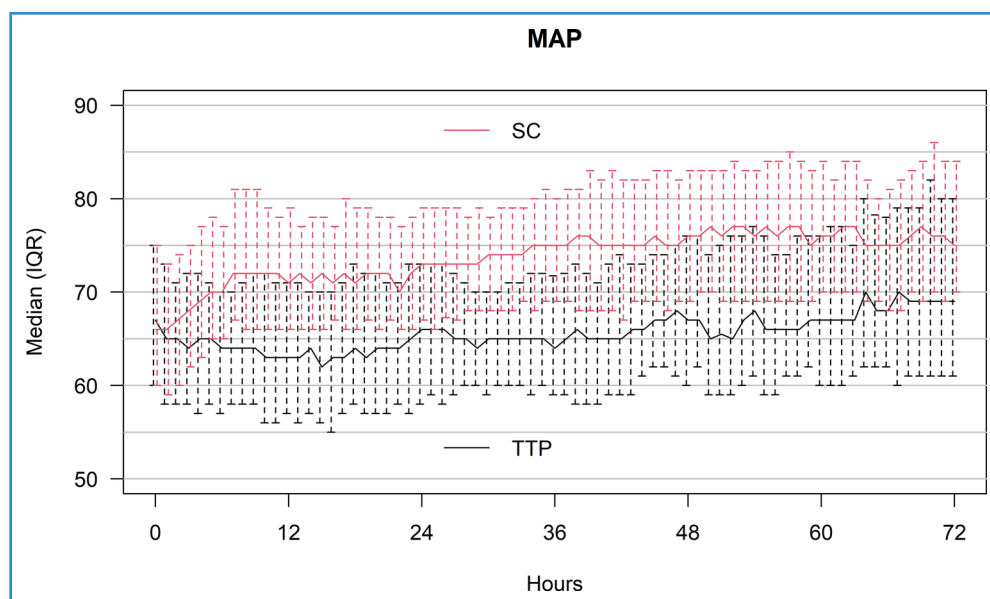


Figure 2. Mean arterial pressure (median [interquartile range (IQR)]) from randomization (0 hr) to 72 hr (targeted tissue perfusion [TTP], standard care [SC]-mean arterial pressure [MAP]-guided; $p < 0.001$ [time]; $p < 0.001$ [group]; and $p = 0.051$ [time-group interaction]).

In addition to deaths and worsening of organ dysfunction, at least one serious adverse reaction was reported in nine patients (5%; six in TTP vs. three in the SC).

DISCUSSION

Hemodynamic support is a main component of treatment of septic shock patients. We compared a strategy aiming at improving tissue perfusion while allowing lower than currently recommended MAP with standard MAP-guided care. The use of TTP did not significantly increase the number of days alive without vasopressors and with normal blood lactate as compared with SC. Hemodynamic problems, interventions, and time patterns of achieved targets were similar in both groups, as were the proportion of adverse events, demonstrating the feasibility of TTP without additional safety concerns.

Treatment of the SC was in broad agreement with international guidelines (1, 2, 4). The patient population fulfilled the Sepsis-3 septic shock definition (3, 4) representing the most severely ill septic shock patients with plausibly the highest likelihood of showing a difference between the trial arms (15). The achieved MAP levels were significantly lower in the TTP vs. SC the group. The observed 24–30% mortality rates at days 30 and 90 were slightly lower than those previously reported (5, 9, 11).

Hemodynamic target recommendations for septic shock treatment are largely based on low-to-moderate quality evidence (1, 2, 4). Targeting MAP of 80–85 mm Hg vs. 65–70 mm Hg did not result in survival benefit but reduced the need for RRT in patients with chronic hypertension (9). The observed MAP levels were substantially higher than targeted (9). A pilot study (OVATION) (10) comparing MAP target of 60–65 mm Hg to MAP target of 75–80 mm Hg found lower mortality in patients older than 75 years

old in the lower MAP target group. The subsequent large 65-trial of patients with vasodilatory shock older than 65 years old, compared MAP target of 60–65 mm Hg with usual care (11). The MAP in usual care group was substantially higher than in the 60–65 mm Hg group (11). Lower MAP target resulted in no significant difference in 90-day mortality. In contrast to SEPSISPAM study (9), no increase in use of RRT was observed in patients with hypertension and lower MAP target (11). A pooled individual-patient data meta-analysis (16) of the SEPSISPAM (9) and OVATION (10) studies and a systematic review and meta-analysis (17) of SEPSISPAM (9), OVATION (10), and 65-trial (11) on MAP targets in septic shock all suggest harm associated with higher exposure to vasopressors and lack of benefits from MAP targets exceeding 65 mm Hg. A recent systematic review and meta-analysis of studies in diverse critically ill patient groups suggested lower mortality associated with lower MAP targets (18).

Peripheral perfusion-guided treatment with 65 mm Hg as the MAP target was applied for 8 hours in the ANDROMEDA-SHOCK trial (5). In a post hoc Bayesian analysis, the posterior probabilities of benefit on mortality using the peripheral perfusion-targeted resuscitation exceeded 90% in several scenarios (6). Peripheral perfusion-targeted therapy also resulted in a higher probability of a lower Sequential Organ Failure Assessment score at 72 hours (6).

TABLE 2.
Primary Composite Outcomes and Secondary Outcomes

Outcomes	Targeted Tissue Perfusion Group (n = 97)	Standard Care Group (n = 97)	Difference in Medians/Proportions (95% CI)	p
Primary outcome				
Days alive in 30 d with normal lactate and without vasopressor or inotropic agents, d, median (IQR)	23 (10–27)	22 (1–27)	0.59 (–3 to 4)	0.418
Secondary outcomes				
Days alive in 30 d, median (IQR)	30 (30–30)	30 (16–30)	1.12 (0.59–2.12) ^b	0.740 ^a
Deaths at day 30, count (%)	24 (24.7)	27 (27.8)	–3.1 (–16.5 to 10.3)	0.745
Time to normal lactate, d, median (IQR)	2 (0–6)	2 (0–5)	0.15 (–2 to 2)	0.917
Days alive with normal lactate in 30 d, median (IQR)	25 (15–29)	25 (8–29)	0.37 (–2 to 4)	0.597
Time to no vasopressor or inotropic agents, d, median (IQR)	3 (1–5)	3 (2–6)	–0.46 (–2 to 0)	0.211
Time to normal lactate and no vasopressor or inotropic agents, median (IQR)	4 (2–9)	5 (2–8)	–0.44 (–2 to 1)	0.553
Days alive in 30 d without vasopressor or inotropic agents, median (IQR)	26 (13–28)	24 (3–27)	1.03 (–2 to 3)	0.239
Days alive without RRT in 30 d, median (IQR)	30 (15–30)	30 (14–30)	–0.23 (–6 to 5)	0.920
Days alive without MV in 30 d, median (IQR)	25 (11–28)	24 (4–28)	0.37 (–3 to 3)	0.626
Days alive without any organ support (RRT, MV) in 30 d, median (IQR)	24 (8–28)	24 (4–28)	0.06 (–4 to 4)	0.959
AKI, n (%)	78 (80)	69 (71)	9.3 (–3.7 to 22.2) ^c	0.180
AKI with RRT, n (%)	35 (36)	31 (32)	4.1 (–10.2 to 18.5) ^c	0.650
Total amount of norepinephrine given in 96 hr, µg/kg, n (%)	766 (819)	947 (1061)	12.2 (–293 to 295)	0.340

AKI = acute kidney injury, IQR = interquartile range, MV = mechanical ventilation, RRT = renal replacement therapy.

^aAdjusted by site and hypertension.

^bCompeting risk analysis, difference as hazard ratio adjusted by site and hypertension.

^cDifference in risk (95% CI).

We allowed substantially lower MAP in the TTP group than any of the previous studies and both the TTP group and the SC group had lower MAP values during vasopressor treatment than intervention and control groups in the previous trials (9–11). Based on systematic reviews and meta-analyses (17, 19), lower than previously recommended MAP targets in shock have recently been suggested (20). The major difference between our trial and the previous ones was assessing markers of tissue perfusion as targets together with allowing lower MAP if tissue perfusion was considered adequate. We cannot determine which of the indicators in TTP was most relevant; this should be addressed in future studies. We cannot exclude the

possibility that indicators of tissue perfusion were also used in the SC. If this was the case, then the main difference in hemodynamic targets between the groups was MAP. We allowed MAP target modification for patients with chronic hypertension and 2-hour test of higher MAP if oliguria was present. Such modifications of MAP targets made the treatment more complicated. The separation of MAP between patients with and without chronic hypertension and the frequent use of temporary increases in MAP during oliguria show that the treatment was feasible. We suggest that the high percentages of patients with chronic hypertension in both groups and their requirement of high doses of norepinephrine may have contributed to

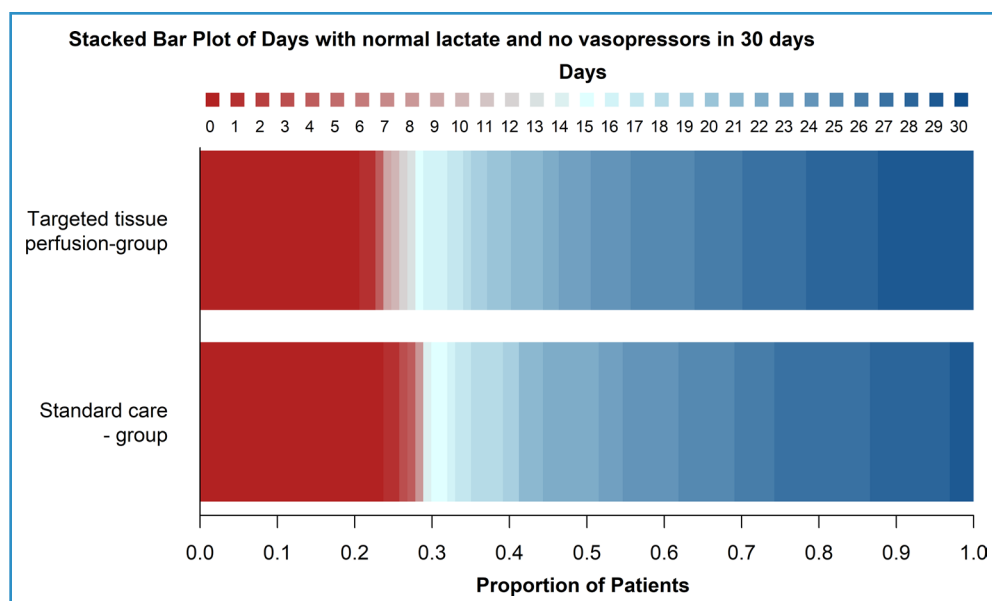


Figure 3. Stacked bar plots of the primary outcome, number of days with normal lactate, and no vasopressors in 30 d as horizontally stacked proportions in each treatment group; red represents worse outcome, blue represents better outcome (interpretation, e.g., in the standard care group 60% of the patients had at least 20 d alive without vasopressors and with normal lactate).

apparently similar cumulative norepinephrine doses in both groups.

As suggested in the editorial (21) accompanying the 65-trial (11), a paradigm change may be needed to avoid unnecessarily high MAP and exposure to vasopressors. We suggest that lower than currently recommended MAP levels should be evaluated in larger clinical trials. Safety measures to protect tissue perfusion, such as those used as targets in this study, are likely necessary. We propose to use somewhat broader MAP targets due to the spontaneous variability of MAP and to define how to adapt vasopressor dose when outside the target.

Our trial has several strengths. First, we recruited patients with septic shock early. Delays in starting the treatment may worsen outcome (22). Second, the clear separation of the observed MAP in study groups including modified MAP targets in the subgroups (hypertension, oliguria) confirm the feasibility of the TTP approach. Third, interventions done to achieve the targets allowed evaluation of protocol compliance.

There are also important limitations to our trial. First, the 6-year trial duration, including interruptions due to COVID-19 pandemic was long. Treatment concepts or patient populations may have changed during the trial. This was the case, for instance, for the Surviving Sepsis Campaign Guidelines with respect

to administration of hydrocortisone. Nevertheless, more than 50% of the patients received glucocorticoids, indicating that new guidelines were followed, at least to a certain extent. Similarly, capillary refill time has been introduced in later versions of the Surviving Sepsis Campaign Guidelines, and it may have influenced clinical decision making among later patients in the SC group. Second, only three sites participated, mainly due to lack of clinical equipoise to lower than recommended MAP targets and complexity of the trial design.

Third, the deferred consent to facilitate early trial inclusion resulted in some early dropouts. Fourth, we only included 17% of all septic shock patients, mainly because of excluding patients with lactate less than 3 mmol/L, those not in early shock, and, to avoid potential harm, those with chronic kidney disease. The inclusion of selected rather than broader patient groups in this phase II trial reduces the generalizability of the results. Fifth, since reaching the TTP targets were only documented for the TTP group, and the clinicians' behavior in how they assessed the patients outside the study protocol was not documented, we cannot exclude the possibility that the main difference in the hemodynamic therapy between the groups was MAP. Even if this were the case, our results suggest that lower than currently recommended MAP levels with concomitant systematic assessment of peripheral tissue perfusion can be applied without obvious increase in adverse events. Our results do not allow evaluation of the importance of specific tissue perfusion variables or their combinations. Sixth, some comorbidities (e.g., diabetes-related microangiopathy and peripheral vascular disease) may impact tissue perfusion and its indicators. We suggest that randomization and the similarity of baseline characteristics of the study groups makes major differences in such confounders unlikely. Seventh, lack of blinding may influence the duration

of vasoactive drug administration. Finally, the primary outcome of this trial has not been externally validated.

CONCLUSIONS

In critically ill patients with septic shock and lactate greater than 3 mmol/L, targeting tissue perfusion while allowing low MAP levels did not increase the number of days alive with normal lactate and without vasopressors or inotropes at 30 days. No additional safety concerns with the TTP strategy were detected compared with SC. Therefore, our findings justify further larger randomized trials in septic shock allowing lower than currently recommended MAP levels with close monitoring of safety measures and adequate tissue perfusion.

ACKNOWLEDGMENTS

We thank Theis Lange for sample size calculations and statistical analysis plan. We are grateful to all trial patients and their relatives, study coordinators, associate investigators, ICU personnel, and treating physicians making this trial possible. We are especially grateful to Mrs. Margaret Lynn Jong-Seiler for her passion for the study and excellent support.

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Drs. Takala, Pettilä, and Jakob were involved in conceptualization. Dr. Selander was involved in formal analysis. Dr. Pettilä and Takala (until August 2018; to Dr. Jakob thereafter) were involved in funding acquisition. All authors except Dr. Selander were involved in investigation. Drs. Takala, Pettilä, Jakob, and Pfortmüller were involved in project administration. Drs. Takala, Pettilä, Jakob, and Pfortmüller were involved in resources. Dr. Pettilä was involved in supervision. Drs. Takala, Pettilä, and Jakob were involved in validation. Dr. Selander was involved in visualization. Dr. Pettilä drafted the first version of the article and finalized it with Drs. Takala, Jakob, Perner, and Selander. All authors have contributed and approved the final version of the article.

The Department of Intensive Care Medicine, Bern University Hospital (Inselspital) has, or has had in the past, research contracts with Orion Corporation, Abbott Nutrition International, B. Braun Medical AG, CSEM SA, Edwards Lifesciences Services GmbH, Kenta Biotech, Maquet Critical Care AB, Omnicare Clinical Research AG, Phagenesis, Nestlé and Cytel, and research & development/consulting contracts with Edwards Lifesciences SA, Maquet Critical Care AB, and Nestlé. The money was paid into a departmental fund, and none of the authors of the department (Drs. Pfortmüller, Jakob, and Takala) received any financial gain. Funding from the Sigrid Juselius Foundation, Instrumentarium Foundation, and Helsinki University Hospital (State funding, VTR-TYH2016243; Finland), the Gottfried & Julia Bangerter-Rhyner Foundation (Switzerland), and Bern University Hospital (Switzerland) was received. The treating doctors/personnel did not receive any money for study patients.

Dr. Selander received support for article research from the National Institutes of Health. The remaining authors have disclosed that they do not have any potential conflicts of interest.

Drs. Jakob and Takala share the senior authorship.

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Trial registration: ClinicalTrials.gov identifier: NCT02579525 (October 19, 2015).

REFERENCES

1. Dellinger RP, Levy MM, Rhodes A, et al; Surviving Sepsis Campaign Guidelines Committee including the Pediatric Subgroup: Surviving sepsis campaign: International guidelines for management of severe sepsis and septic shock: 2012. *Crit Care Med* 2013; 41:580–637
2. Rhodes A, Evans LE, Alhazzani W, et al: Surviving sepsis campaign: International guidelines for management of sepsis and septic shock: 2016. *Crit Care Med* 2017; 45:486–552
3. Singer M, Deutschman CS, Seymour CW, et al: The third international consensus definitions for sepsis and septic shock (Sepsis-3). *JAMA* 2016; 315:801–810
4. Evans L, Rhodes A, Alhazzani W, et al: Executive summary: Surviving sepsis campaign: International guidelines for the

- management of sepsis and septic shock 2021. *Crit Care Med* 2021; 49:1974–1982
5. Hernández G, Ospina-Tascón GA, Damiani LP, et al; The ANDROMEDA SHOCK Investigators and the Latin America Intensive Care Network (LIVEN): Effect of a resuscitation strategy targeting peripheral perfusion status vs serum lactate levels on 28-day mortality among patients with septic shock the ANDROMEDA-SHOCK randomized clinical trial. *JAMA* 2019; 321:654–664
 6. Zampieri FG, Damiani LP, Bakker J, et al: Effects of a resuscitation strategy targeting peripheral perfusion status versus serum lactate levels among patients with septic shock. A Bayesian reanalysis of the ANDROMEDA-SHOCK trial. *Am J Respir Crit Care Med* 2020; 201:423–429
 7. Nguyen HB, Rivers EP, Knoblich BP, et al: Early lactate clearance is associated with improved outcome in severe sepsis and septic shock. *Crit Care Med* 2004; 32:1637–1642
 8. Nguyen HB, Loomba M, Yang JJ, et al: Early lactate clearance is associated with biomarkers of inflammation, coagulation, apoptosis, organ dysfunction and mortality in severe sepsis and septic shock. *J Inflamm (Lond)* 2010; 7:6
 9. Asfar P, Meziani F, Hamel JF, et al; SEPSISPAM Investigators: High versus low blood-pressure target in patients with septic shock. *N Engl J Med* 2014; 370:1583–1593
 10. Lamontagne F, Meade MO, Hébert PC, et al; Canadian Critical Care Trials Group: Higher versus lower blood pressure targets for vasopressor therapy in shock: A multicentre pilot randomized controlled trial. *Intensive Care Med* 2016; 42:542–550
 11. Lamontagne F, Richards-Belle A, Thomas K, et al; 65 trial investigators: Effect of reduced exposure to vasopressors on 90-day mortality in older critically ill patients with vasodilatory hypotension: A randomized clinical trial. *JAMA* 2020; 323:938–949
 12. Pettilä V, Merz T, Wilkman E, et al: Targeted tissue perfusion versus macrocirculation-guided standard care in patients with septic shock (TARTARE-2S): Study protocol and statistical analysis plan for a randomized controlled trial. *Trials* 2016; 17:384
 13. Sacerdoti D, Gaiani S, Buonamico P, et al: Interobserver and interequipment variability of hepatic, splenic, and renal arterial Doppler resistance indices in normal subjects and patients with cirrhosis. *J Hepatol* 1997; 27:986–992
 14. KDIGO clinical practice guideline for acute kidney injury. *Kidney Int Suppl* 2012; 2:1–138
 15. Iwashyna TJ, Burke JF, Sussman JB, et al: Implications of heterogeneity of treatment effect for reporting and analysis of randomized trials in critical care. *Am J Respir Crit Care Med* 2015; 192:1045–1051
 16. Lamontagne F, Day AG, Meade MO, et al: Pooled analysis of higher versus lower blood pressure targets for vasopressor therapy septic and vasodilatory shock. *Intensive Care Med* 2018; 44:12–21
 17. Richards-Belle A, Hylands M, Muttalib F, et al: Lower versus higher exposure to vasopressor therapy in vasodilatory hypotension: A systematic review with meta-analysis. *Crit Care Med* 2023; 51:254–266
 18. D'Amico F, Pruna A, Putowski Z, et al: Low versus high blood pressure targets in critically ill and surgical patients: A systematic review and meta-analysis of randomized controlled trials. *Crit Care Med* 2024; 52:1427–1438
 19. Carayannopoulos KL, Pidutti A, Upadhyaya Y, et al: Mean arterial pressure targets and patient-important outcomes in critically ill adults: A systematic review and meta-analysis of randomized trials. *Crit Care Med* 2023; 51:241–253
 20. Roberts RJ, Chen JT, Sevransky JE, et al: Under pressure: Do we “dare change our way of caring” for patients with shock? *Crit Care Med* 2023; 51:326–328
 21. Marshall JC: Choosing the best blood pressure target for vasopressor therapy. *JAMA* 2020; 323:931–933
 22. Roberts RJ, Miano TA, Hammond DA, et al; Observation of VariatiOn in fLUids adMinistEred in shock-CHaracterizAtion of vaSoprEссор Requirements in Shock (VOLUME-CHASERS) Study Group and SCCM Discovery Network: Evaluation of vasopressor exposure and mortality in patients with septic shock. *Crit Care Med* 2020; 48:1445–1453